GCLOUD AND CLOUDSHELL

GOOGLE CLOUD PLATFORM

Gcloud

- **Command line interface** to interact with Google Cloud Resources
- Most GCP services can be managed from CLI using Gcloud:
  - Compute Engine Virtual Machines
  - Managed Instance Groups
  - Databases
  - and ... many more
- You can create/delete/update/read existing resources and perform actions like deployments as well!
- (REMEMBER) SOME GCP services have specific CLI tools:
  - Cloud Storage - gsutil
  - Cloud BigQuery - bq
  - Cloud Bigtable - cbt
  - Kubernetes - kubectl (in addition to Gcloud which is used to manage clusters)

Cloud Shell

Gcloud - Getting Started

Installation

- Gcloud is part of Google Cloud SDK
  - Cloud SDK requires Python
  - Instructions to install Cloud SDK (and Gcloud) => ***https://cloud.google.com/sdk/docs/install***
- You can also use Gcloud on Cloud Shell

Connecting to GCP

- **`gcloud init`** - initialize or reinitialize gcloud
  - Authorize gcloud to use your user account credentials
  - Setup configuration
    - Includes current project, default zone etc
- **`gcloud config list`** - lists all properties of the active configuration

gcloud config set

- Sets the specified property in your active configuration
  - **`gcloud config set core/project VALUE`**
  - **`gcloud config set compute/region VALUE`**
  - **`gcloud config set compute/zone VALUE`**
  - **`gcloud config set core/verbosity VALUE(debug)`**
- Syntax - **`gcloud config set SECTION/PROPERTY VALUE`**
  - **`core`**, **`compute`** - SECTIONS
  - **`project`**, **`region`**, **`zone`** - PROPERTIES
  - Specifying **`core`** is optional as it is the default SECTION!
    - **`gcloud config set project VALUE`**
    - **`gcloud config set verbosity VALUE(debug)`**
  - Get more details with **`gcloud config set --help`**
    - Look for `AVAILABLE PROPERTIES` in the content
- Opposite - **`gcloud config unset`**

Cloud Shell

Playing with gcloud config set

- gcloud config set compute/region us-east2
- gcloud config set compute/zone us-east1-b
- gcloud config list

```
testing@cloudshell:~ (useful-device-303710)$ gcloud config list
[component_manager]
disable_update_check = True
[compute]
gce_metadata_read_timeout_sec = 30
region = us-east1
zone = us-east1-b
[core]
account = testing@gmail.com
disable_usage_reporting = True
project = useful-device-303710
verbosity = info
[metrics]
environment = devshell
```

Gcloud - Managing Multiple Configurations

- Scenario: You are working on multiple projects from the same machine. You would want to be able to execute commands using different configurations.
  - How do you simplify this?
    - **`gcloud config configurations create/delete/describe/activate/list`**
      - Create new configuration: **`gcloud config configurations create NAME(dev)`**
      - Activate specific configuration: **`gcloud config configurations activate NAME(dev)`**
      - List Configurations: **`gcloud config configurations list`**

Cloud Shell

gcloud command structure - Playing with Services

- **`gcloud GROUP SUBGROUP ACTION ...`**
  - GROUP - config or compute or container or dataflow or functions or iam or ..
    - Which service group are you playing with?
  - SUBGROUP - instances or images or instance-templates or machine-types or regions or zones
    - Which sub group of the service do you want to play with?
  - ACTION - create or list or start or stop or describe or ...
    - What do you want to do?
- **Examples:**
  - gcloud compute instances list
  - gcloud compute zones list
  - gcloud compute regions list
  - gcloud compute machine-types list
  - gcloud compute machine-types list --filter="zone:us-central1-b"
  - gcloud compute machine-types list --filter="zone:(us-central1-b europe-west1-d)"

gcloud compute instances create

- Creating Compute Instances
  - **`gcloud compute instances create [NAME]`**
    - Options:
      - --machine-type (default type is n1-standard-1 - gcloud compute machine-types list)
      - --custom-cpu --custom-memory --custom-vm-type(n1/n2) (Custom Machine)
        - --custom-cpu 6 --custom-memory 3072MB --custom-vm-type n2
      - --image or --image-family or --source-snapshot or --source-instance-template or --source-machine-image (beta)
      - --service-account or --no-service-account
      - --zone=us-central1-b
      - --tags (List of tags - Allow network firewall rules and routes to be applied to VM instances)
      - --preemptible
      - --restart-on-failure(default) --no-restart-on-failure --maintenance-policy(MIGRATE(default)/TERMINATE)
      - --boot-disk-size, --boot-disk-type --boot-disk-auto-delete(default) --no-boot-disk-auto-delete
      - --deletion-protection --no-deletion-protection(default)
      - --metadata/metadata-from-file startup-script/startup-script-url
        - --metadata-from-file startup-script=/local/path/to/script/startup OR --metadata startup-script="echo 'hello world'"
        - shutdown-script
      - --network --subnet --network-tier (PREMIUM (default), STANDARD)
      - --accelerator="type=nvidia-tesla-v100,count=8" --metadata="install-nvidia-driver=True" (GPU)

Compute Instances - Default Region and Zone

- Three Options:
  - Option 1 (Centralized Configuration): gcloud compute project-info add-metadata
    - --metadata=[google-compute-default-region=REGION | google-compute-default-zone=ZONE]
  - Option 2 (Local gcloud configuration): gcloud config set compute/region REGION
  - Option 3 (Command Specific): --zone or --region in the command
- Priority: Option 3 (if exists) overrides Option 2 (if exists) overrides Option 1

Cloud Shell

List and Describe commands

- Typically list commands are used to list a set of resources
  - gcloud compute RESOURCES list
    - gcloud compute images/regions/zones/disk-types list
    - gcloud compute instances/disks/snapshots list
  - Most list commands support a few common options
    - --filter="zone:VALUE"
    - --sort-by (NAME, ~NAME)
    - --uri (***https://www.googleapis.com/compute/v1/projects/windows-sql-cloud/global/images/sql-2019-web-windows-2019-dc-v20210112***)
    - gcloud compute images list --sort-by NAME --filter "PROJECT:(windows-cloud ubuntu-os-cloud)"
- Typically describe commands are used to describe a specific resource
  - gcloud compute images describe ubuntu-1604-xenial- v20210203 --project ubuntu-os-cloud
  - gcloud compute regions describe us-central1

Cloud Shell

Playing with Compute Instances - gcloud

- Playing with compute instances
  - gcloud compute instances list/start/stop/delete/reset/describe/move
    - gcloud compute instances start example-instance
    - gcloud compute instances stop example-instance-1 example-instance-2
    - gcloud compute instances delete example-instance
      - --delete-disks=VALUE (all or data or boot)
      - --keep-disks=VALUE (all or data or boot)
    - gcloud compute instances move example-instance-1 --zone us-central1-b --destination-zone us-central1-f
      - Move a VM from one zone to another

Cloud Shell

Playing with Instance Templates

- gcloud compute instance-templates create/delete/describe/list
  - gcloud compute instance-templates create INSTANCE-TEMPLATE
    - --source-instance=SOURCE_INSTANCE --source-instance-zone (Which instance to create a template from?)
    - Supports almost all options supported by **`gcloud compute instances create [NAME]`**
      - --image or --image-family or --source-snapshot or --source-instance-template
      - --service-account or --no-service-account
      - --tags
      - --preemptible
      - --restart-on-failure(default) --no-restart-on-failure --maintenance-policy(MIGRATE(default)/TERMINATE)
      - --boot-disk-size, --boot-disk-type --boot-disk-auto-delete(default) --no-boot-disk-auto-delete
      - --deletion-protection --no-deletion-protection(default)
      - --metadata/metadata-from-file startup-script/startup-script-url
      - --network --subnet --network-tier (PREMIUM (default), STANDARD)
      - --accelerator="type=nvidia-tesla-v100,count=8" --metadata="install-nvidia-driver=True" (GPU)
  - Using Instance Tempate to create an instance
    - gcloud compute instances create my-test-vm --sourc e-instance-template=my-instance-template-with-custom-image